Team 02: The Perks of Being a First Mover

Does White's Advantage in Chess Grow or Shrink with Rating?

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Repository: `github.com/Sahil-M-Deo/The-Perks-of-Being-a-First-Mover`

1. Main Hypothesis / Question

Chess is a perfect-information game where White enjoys a built-in first-mover advantage. Conventional wisdom states that moving first dictates the opening initiative. However, we ask whether this advantage scales linearly with player skill.

Our main hypothesis is that White's win rate (the statistical manifestation of the first-mover advantage) is not constant across skill levels. We hypothesize that the advantage peaks at intermediate Elo ratings due to players being skilled enough to exploit opening traps but lacking rigorous defensive theory, and the advantage shrinks significantly at the Master and Grandmaster level (2200+ Elo), where Black possesses the theoretical knowledge to perfectly neutralize the opening initiative and equalize the position.

To test this, we will utilize the Lichess Open Database. We will download a single month's standard chess PGN file (containing millions of games). By parsing the PGN headers, we will extract White's Elo rating, Black's Elo rating, and the match outcome (1-0, 0-1, 1/2-1/2). To ensure fair comparisons, we will filter out matches with a rating disparity greater than 50 points. We will then group the remaining millions of games into fixed Elo brackets (e.g., 1000-1200, 1500-1700, 2200+) and calculate the win, loss, and draw distributions for White in each bracket. By analyzing this empirical big data, we will determine if mastery of the board ultimately erases the first-mover advantage.

1.1 Sources

- **Lichess Open Database:** database.lichess.org (Standard Chess PGNs).
- **FIDE Elo Rating System Mathematics:** The logistic curve and expected win probability formulas used to calculate chess ratings.

1.2 Programming languages and tools

C++ for writing a highly optimized parsing routine to quickly extract headers from multi-gigabyte PGN files. Python (pandas, scipy.stats, statsmodels, matplotlib) in Jupyter notebooks for bucketing the Elo ratings and calculating statistical significance. Plotly for an interactive dashboard visualizing win/draw rates across Elo brackets. GitHub for version control and sharing.

Statistical questions we expect to answer: How to handle massive sample sizes where even trivial differences might trigger statistical significance, how to properly model the massive spike in draw rates at higher Elos, and determining if the shift in White's win rate is mathematically significant or just variance.

1.3 Datasets

Dataset	Use	Access
Lichess Standard Games PGN	Primary dataset: White Elo, Black Elo, and Match Outcome.	database.lichess.org (Free)
FIDE Rating Distributions	Baseline comparison for expected draw rates at elite levels.	ratings.fide.com (Public)

2. Logistics

2.1 Spreadsheet

Updated project progress sheet: `https://docs.google.com/spreadsheets/d/1pvTllHR4a1Sl7GHtjKnvLeb-z` `edit?usp=sharing`. We have added our planned dates for each stage and rows for the GitHub repository, main milestones, and member roles.

2.2 Work breakup and plan

Member	Responsibility
Sanidhya	Data Engineering: Write C++ routines to parse massive PGN files, extract specific Elo/Outcome headers, and export clean CSVs.
Daksh Maahor	Data Cleaning: Filter out games with >50 Elo delta, handle missing data, and structure the Elo brackets.
Sahil	Statistical Analysis: Run statistical significance tests on the brackets and analyze draw rates.
VadaPavJai	Visualization: Build the Plotly dashboard mapping White win/draw rates against Elo, and format the final report.

Timeline:

- **By 19 Oct:** Download Lichess PGN dataset and finalize the C++ parsing script.
- **Checkpoint 1 (21 Oct):** Millions of games successfully parsed, filtered by Elo delta, and exported as a clean dataframe. Initial win/loss/draw ratios calculated per bracket.
- **22-31 Oct:** Run statistical tests (e.g., chi-square tests across brackets), analyze the impact of the draw rate at high Elos.
- **Checkpoint 2 (4 Nov):** The final set of deliverables must be handed in. This includes robustness checks (ensuring time controls like Blitz vs Classical don't skew the data), completed Plotly dashboard, and the final report.

Working style: GitHub repo as the single source of truth (issues for tasks, pull requests reviewed by one other member).

3. Help Needed from the Teaching Team

- **Big Data Optimization:** Advice on processing and chunking multi-gigabyte text (PGN) files efficiently if our local machines face memory limits.
- **Statistical Modeling:** Guidance on how to treat the "Draw" outcome in our statistical tests, as draw rates heavily correlate with Elo and complicate binary win/loss models.